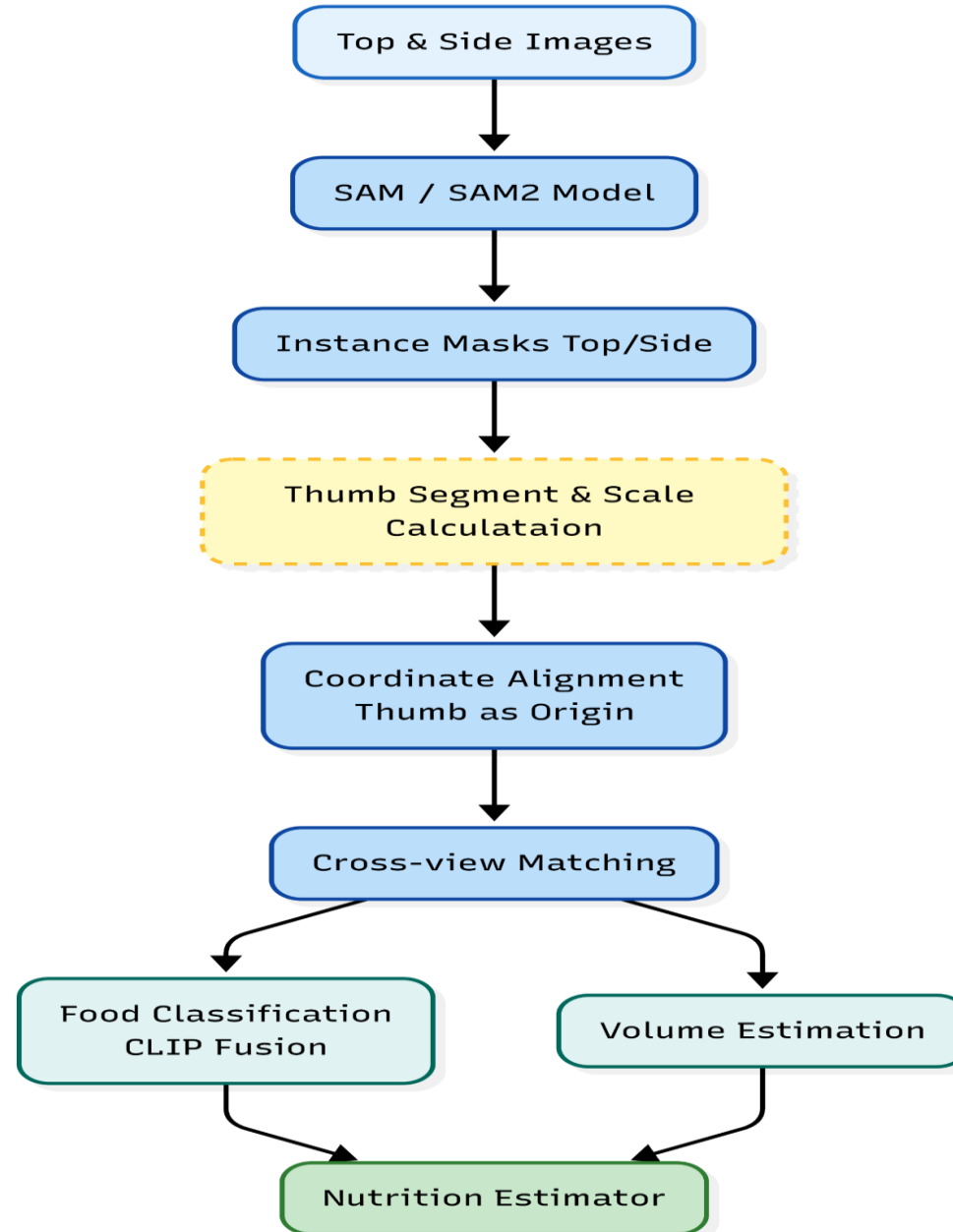


Food & Calorie Intake Tracking System

System Architecture

Pipeline Diagram:



Sequential Inference Pipeline

Step-by-step processing flow :

- 1. Input:** Top & Side Image Acquisition
- 2. Segmentation:** SAM2 / SAM-HQ extracts per-food instance masks from images. Enables isolation of each food object.
- 3. Calibration:** Thumb detected using YOLO/SAM fine-tune. Used as metric reference to convert pixels into real-world scale.
- 4. Alignment:** Coordinate Alignment setting Thumb as origin
- 5. Matching:** Matches same food across top and side views using graph matching + Hungarian algorithm.
- 6. Classification:** Uses CLIP ViT-L fused with Food-101 CNN to recognize food categories per instance.
- 7. Volume Estimation:** Combines geometry from top & side masks and MLP regression to estimate food volume.
- 8. Nutrition Estimation:** Maps volume → mass → calories using nutrition database. Produces protein, fat, carb values.

Technology Component Selection:

Component	Selected Technology
Segmentation	SAM2 / SAM-HQ
Reference Detection	YOLOv8-nano or SAM fine-tune
Classification	CLIP ViT-L + Food-101 CNN
Instance Matching	Graph Matching + Hungarian Alg.
Volume Estimation	Hybrid Geometry + MLP